FIFTH SEMESTER (CUFYUGP) DEGREE EXAMINATION: MODEL PAPER 3

B.Sc. PHYSICS HONOURS

PROPERTIES OF SOLIDS (SOLID STATE PHYSICS I)

Module III (Units 13–16) and Module IV (Unit 17): Semiconductor Properties; Types of Magnetism

Maximum Time: 2 Hours		**Maximum Marks: 70**
	Section A	
	All Questions can be answered. Each Question carries 3 marks (Ceiling: 24 Marks)	
1	Define drift velocity and mobility of a charge carrier. Which has the greater mobility in a semiconductor, an electron or a hole?	**3**
2	Explain the concepts of drift current and diffusion current. How are they different?	**3**
3	Obtain an expression for the electrical conductivity of an intrinsic semiconductor in terms of the intrinsic carrier concentration n_i and the mobilities μ_n and μ_p.	**3**
4	How does the conductivity of an intrinsic semiconductor vary with temperature? How can the band gap be determined from a plot of $\ln \sigma$ against $1/T$?	**3**
5	An intrinsic germanium crystal has a hole density of $10^{19}\ \mathrm{m^{-3}}$ at room temperature. When doped with antimony, the hole density decreases to $10^{17}\ \mathrm{m^{-3}}$ at the same temperature. Calculate the majority carrier density.	**3**
6	State the law of mass action. For what type of semiconductors does it hold? Can a semiconductor containing both n-type and p-type impurities behave as an intrinsic semiconductor? Give the reason.	**3**
7	Prove that the Fermi level in an intrinsic semiconductor lies in the middle of the forbidden gap if the effective masses of electrons and holes are equal.	**3**
8	The conductivity of an n-type germanium semiconductor is $39\ \Omega^{-1}\mathrm{m}^{-1}$. If the mobility of electrons in germanium is $0.39\ \mathrm{m^2\,V^{-1}s^{-1}}$, find the concentration of the donor atoms.	**3**
9	Define magnetisation, magnetic susceptibility and relative permeability of a medium. How are the susceptibility and the relative permeability related?	**3**
10	Name the different types of magnetism observed in solids. How do orbital and spin motions of electrons give rise to permanent magnetic moments in atoms?	**3**
	Section B	
	All Questions can be answered. Each Question carries 6 marks (Ceiling: 36 Marks)	
11	Derive an expression for the density of electrons in the conduction band of an n-type semiconductor at moderate temperatures. How does the electron concentration depend on the donor concentration?	**6**
12	Show a typical variation of conductivity with temperature for an extrinsic (n-type) semiconductor and explain the different regions of the curve.	**6**
13	Show that the product of the electron and hole concentrations in a semiconductor is constant at a given temperature. How is the energy gap determined from the measurement of the electrical conductivity of a semiconductor?	**6**
14	The electron and hole mobilities in a Si sample are 0.135 and $0.048\ \mathrm{m^2/V\text{-}s}$ respectively. Determine the conductivity of intrinsic Si at $300\ \mathrm{K}$ if the intrinsic carrier concentration is $1.5 \times 10^{16}\ \mathrm{m^{-3}}$. The sample is then doped with 10^{23} phosphorus atoms$/\mathrm{m^3}$. Determine the equilibrium hole concentration, the conductivity, and the position of the Fermi level relative to the intrinsic level.	**6**
15	In intrinsic GaAs, the electron and hole mobilities are 0.85 and $0.04\ \mathrm{m^2/V\text{-}s}$ respectively, and the corresponding effective masses are $0.068\,m_0$ and $0.5\,m_0$, where m_0 is the rest mass of an electron. Given the energy band gap at $300\ \mathrm{K}$ as $1.43\ \mathrm{eV}$, determine the intrinsic carrier concentration and the conductivity.	**6**
16	(a) In an n-type semiconductor the Fermi level lies $0.5\ \mathrm{eV}$ below the conduction band. If the concentration of donor atoms is tripled, find the new position of the Fermi level, given $kT = 0.03\ \mathrm{eV}$. (b) The conductivity of intrinsic Si is 4.17×10^{-5} and $4 \times 10^{-4}\ (\Omega\,\mathrm{m})^{-1}$ at $0°\mathrm{C}$ and $27°\mathrm{C}$ respectively. Determine the average band gap of Si.	**6**

17	Distinguish between the characteristic features of diamagnetism, paramagnetism, ferromagnetism, antiferromagnetism and ferrimagnetism. Give an example of each type of material. Comment on the temperature variation of susceptibility for each type.	**6**
18	Derive the expression for the position of the Fermi level in a p-type semiconductor at moderate temperatures. How does the Fermi level of a p-type semiconductor change with temperature?	**6**
Section C		
Answer any ONE. Each Question carries 10 marks (1x10=10 Marks)		
19	Define drift velocity and mobility. Obtain an expression for the conductivity of an intrinsic semiconductor and discuss its variation with temperature. Extend the result to n-type and p-type semiconductors, and explain how the conductivity changes with impurity content.	**10**
20	Derive expressions for the electron and hole concentrations in an intrinsic semiconductor in terms of the Fermi energy. Use these results to obtain the intrinsic carrier concentration, and show that the Fermi level lies halfway between the valence band and the conduction band.	**10**